Dakyung Kang

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EDUCATION

University of Utah Salt Lake City, UT
B.S. in Electrical and Computer Engineering *Dec 2025*

PUBLICATION

H. Shin, **D. Kang**, J. Kim, and N. Kim, "CMOS Impedance Sensor with 128×128 Microelectrode Arrays for Measuring Surface Coverage of Microparticles," *2025 IEEE SENSORS Conference*, Vancouver, Canada, accepted for publication, Oct. 2025.

SKILLS

Programming Languages: C/C++, Python.
Hardware Description: Verilog, SystemVerilog, RTL Design.
EDA & Tools: Cadence (Innovus, Genus, Virtuoso), Calibre, MATLAB.

EXPERIENCE

Patent Engineer Seoul, Republic of Korea
Honesty & JR Partners *Feb 2026 – Jun 2026*

- Analyzed circuit operating principles of power semiconductor and display circuits at the transistor level based on inventor disclosure forms.
- Conducted prior art searches for 5 patent proposals across EV inverter circuits, MOSFET fabrication, identifying 3 reference patents per case and drafted patent specification and claims for display fabrication and display share/private mode circuits.

Lab Assistant Salt Lake City, UT
Biomedical Micro-Nano Systems Lab *Aug 2024 – Dec 2025*

- Validated sensing performance of **CMOS electrode array** by correlating impedance with optical microscopy across five microparticle concentrations.
- Measured impedance across electrode spacings (3μm vs 4μm, 20–100kHz), achieving 1.63x higher SNR with a low spacing gap and adopting it for subsequent immunoassay measurements.
- Analyzed $\Delta|Z|$ via heatmap visualization in **MATLAB** by mapping results to surface coverage in 0.25% to 98.65% and confirmed spatial correlation with microparticle distribution.

TECHNICAL PROJECTS

High Speed Data Acquisition Controller *Sep 2025 – Jun 2026*

- Designed an 8-channel biosignal acquisition controller (ECG/EEG/EMG) in **SystemVerilog** for a bedside patient monitor, featuring Round-Robin, Priority, Weighted, and Dynamic arbitration to prioritize ECG sampling and a 16-entry FIFO with 2-level watermark status to decouple ADC sampling from SPI output.
- Built testbenches with 17 SystemVerilog Assertions and constrained-random stimulus covering FIFO overflow/underflow, simultaneous read/write, all 16 arbitration mode transitions, reset-during-operation, and all-8-channel contention, achieving 100% functional coverage via **Questa covergroups**.
- Integrated **8x4 Mac array** for real-time biosignal detection and tested against a **Python golden model** over 1,000 random inputs with 0 mismatches.

DES Processor – VLSI *Sep 2025 – Dec 2025*

- Executed a full RTL-to-GDSII implementation flow for a 16-round Feistel DES engine using **Verilog** in a **TSMC 180nm** node.
- Resolved critical synthesis errors in **Genus** by redesigning tristate logic into synchronous register-based outputs and passed standard cell library compatibility.
- Achieved timing closure in **Innovus** by improving WNS from -12.742ns with 803 timing violations to +4.315ns through H-tree topology Clock Tree Synthesis.
- Ensured manufacturability with 0 DRC/LVS violations using **Calibre**, including automatic insertion of 17 antenna protection diodes via **Innovus** to resolve antenna effects.